

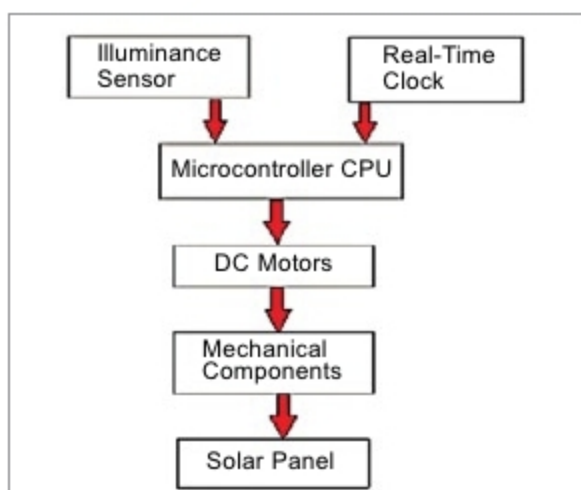


Fig. 3: Block diagram of an active solar tracking system

ously detect and track incident light in 3D space at a broad range of ambient temperatures with high accuracy and fast response, without auxiliary power supply or human intervention. It has been demonstrated that artificial phototropism can achieve up to 400 per cent energy-harvesting enhancement over conventional non-tropistic materials under oblique illumination.

Importance of solar energy

Solar cells turn sunlight into electricity. This technology offers a more Earth-friendly way to produce power than burning coal and other fossil fuels. Importance of solar energy has increased every year. World population and energy needs have increased linearly. At present, total energy requirement of the world is about thirteen terawatt (TW). Total renewable energy source of the world under various heads is as follows: 0.5TW hydro-electric energy, 2TW to 4TW wind energy, 12TW geothermal energy, 2TW ocean wave energy and a stunning 120,000TW solar energy. This means we must tap this source efficiently for our increasing energy demands.

Present-day commercial silicon solar cells have about 22 per cent efficiency. One of the most efficient ways to increase photovoltaic performance is integrating heliotropism, that is, the ability to track the Sun across the sky, using solar cells for more efficient photon collection. Most active heliotropic systems that are used with solar panels have external positioning panels with motors, illuminance sensors, microcontroller CPUs, GPS tracking devices, etc. These systems are costly and need regular maintenance and power supply for operation.

The block diagram of an active

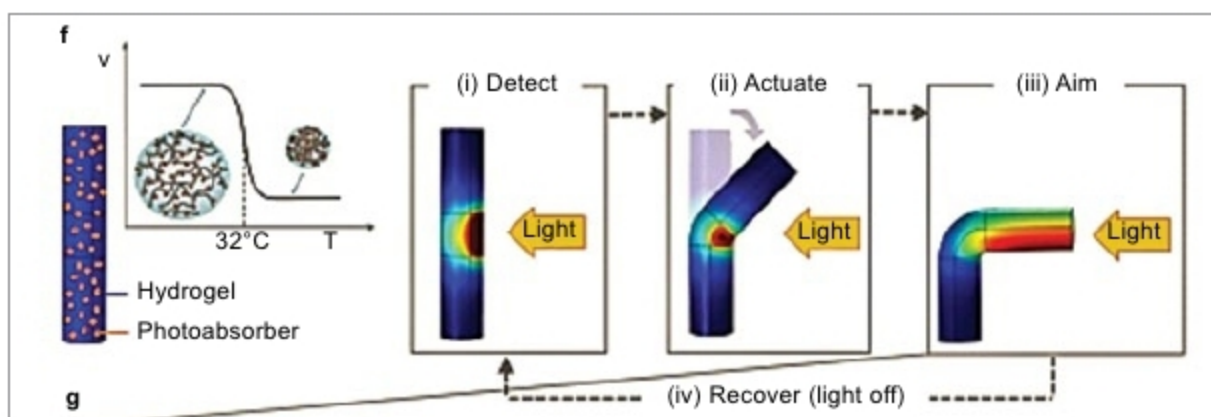


Fig. 4: Mechanism of phototracking of SunBOT

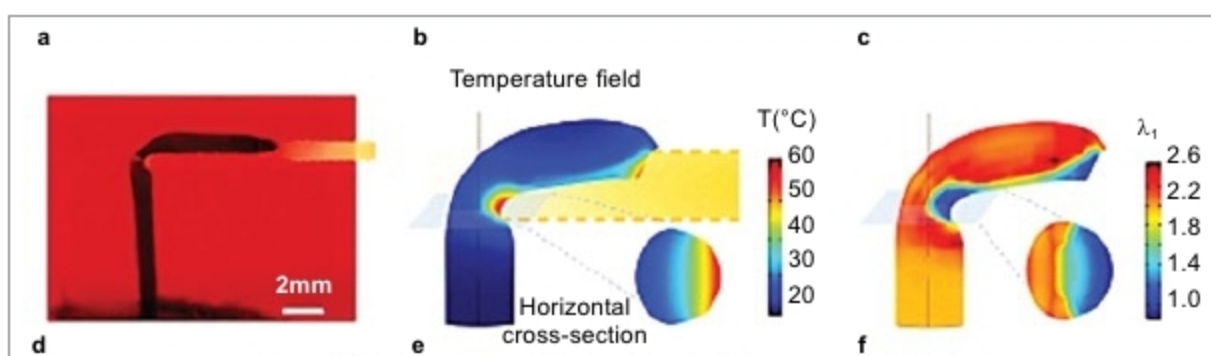


Fig. 5: Temperature distribution and stretch field at steady aiming state

heliotropic system is shown in Fig. 3. Integrating heliotropic property into the photovoltaic system renders the system passive, reducing extraneous parts. No external power supply is required because it is light-driven. It is very cost-effective and requires minimum maintenance.

Design and operating principle of SunBOT

To mimic biological asymmetric growth, reversible photo-thermo-responsive soft materials that shrink on illumination are used to fabricate artificial stem-like cylindrical pillars. Like stems, SunBOT pillars bend towards stimuli due to asymmetric deformation between the illuminated high-temperature region and the shaded low-temperature region.

SunBOT can be constructed from many reversible thermo-responsive soft polymer materials like hydrogels, liquid crystal elastomers (LCEs) and azobenzene/spirocyan for mechanical actuation. Photo-thermally-responsive SunBOT materials like hydrogels are mixed homogeneously with nano-photo-absorbers, either gold nanoparticles or reduced graphene-oxide, which are all photo-responsive. The homogeneous mixture is packed in a polyurethane mesh for structural support.

These nanoparticles absorb light via plasmonic resonance modes and generate heat that changes volume

of the thermo-responsive polymer in response to temperature change. The system has a low actuation temperature of 32°C. Reaction time is less than ten seconds and is tunable. One of the thermo-responsive smart hydrogels that is used often along with photo-responsive nanomaterial for construction of SunBOT is poly (N-isopropylacrylamide), that is, PNIPAAm, to create a system with tunable response time by varying concentration of nanoparticles.

Similar to sunflower, the phototracking process involves four steps: detection, actuation, aiming and recovery, as shown in Fig. 4. On illumination, SunBOT detects incident light via surface plasmon resonance of the incorporated nano-photo-absorbers and actuates towards the source due to local heating and asymmetric deformation.

Asymmetric deformation is induced by establishing an appropriate temperature gradient across the pillar, where illuminated front-side temperature is more than the hydrogel's lower critical solution temperature (LCST) and the shaded back-side temperature (T_b) remains below LCST. Hence, illuminated side shrinks.

This illuminated side shrinkage results in a bending effect similar to that observed in plant stem, as shown in Fig. 5. The figure also shows temperature distribution and stretch field of SunBOT at steady aiming state. As soon as SunBOT aims at the light source, it